JOURNAL OF ADVANCED TRANSPORTATION
J. Adv. Transp. 2016; **50**:1520–1530
Published online 13 September 2016 in Wiley Online Library (wileyonlinelibrary.com). DOI: 10.1002/atr.1414

Development of a driving cycle for Colombo, Sri Lanka: an economical approach for developing countries

Uditha Galgamuwa*, Loshaka Perera and Saman Bandara

Transportation Engineering Division, Department of Civil Engineering, University of Moratuwa, Moratuwa, Sri Lanka

SUMMARY

Financial constraints and lack of availability of traffic-related information significantly hinder the development of driving cycles in developing countries. This paper proposes an economical, practical, accurate methodology for the development of driving cycles, including the development of a driving cycle for Colombo, Sri Lanka. The proposed methodology captures regional traffic and road conditions and selects a model that represents the collected data sample with minimum available traffic-related information. Existing methods were modified for route selection by dividing routes into links using nodes or physical junctions to minimize the number of trips required for data collection. Speed–time data for respective links were used to reconstruct speed–time profiles of identified origin–destination pairs. The on-board method was used for data collection, and the Markov chain theory was used to develop a transition probability matrix of state changes. An additional matrix was introduced to the existing method to improve model representativeness to the collected data sample. Copyright © 2016 John Wiley & Sons, Ltd.

KEY WORDS: driving cycle; route selection; cycle construction; data validation; cycle validation; vehicular emission

1. INTRODUCTION

The vehicle population in Sri Lanka is growing exponentially. According to the Department of Motor Traffic, the total vehicle population in Sri Lanka has increased by 80%, and the number of cars on Sri Lankan roads has increased by more than 50% from the year 2007 to 2014 [1]. Colombo, the capital of Sri Lanka, is affected by the increased vehicle usage, which causes a consistent rise in air pollution levels and negative environmental impacts. Therefore, emission standards should be established for Colombo, using local traffic behavior.

Outdated vehicle emission testing (VET) in Sri Lanka is based on emission levels at idling and 2500 revolutions per minute (rpm) without any load [2] even though the emission inventories should measure based on traffic behavior in Sri Lanka. The driving cycle method was proposed for estimating emission inventories in Sri Lanka, and an initial driving cycle was developed in 2013 based on data collected from one road segment [3]. However, the initial driving cycle was developed to understand possible methods of driving cycle development, revealing the need for an accurate driving cycle with a limited budget. The objective of this research was to develop an economical, precise driving cycle for light vehicles in Sri Lanka in order to promote the method in countries with limited budgets and resources.

2. LITERATURE REVIEW

Air quality is one of the major factors for quality of life in residential areas in urban cities [4]. Vehicular emission has been identified as a primary source of urban air pollution [5], thereby placing a

*Correspondence to: Uditha Galgamuwa, Transportation Engineering Division, Department of Civil Engineering, University of Moratuwa, Moratuwa, Sri Lanka. E-mail: nandun8781@gmail.com

significant responsibility for air quality on the transportation sector [6]. Efficient estimation of emission inventories should be conducted using available resources, knowledge and budgets. Driving cycle is one of the major methods which have been used around the world for decades to estimate emission inventories in a given region or a road segment. A driving cycle is a series of data points that represents speed versus time in a specific region or part of a road segment [7]. It is a synthesized driving pattern to represent average real-world driving [8]. Driving cycles such as FTP72, FTP75, LA92 [9] and LA01 in the USA [10]; the Sydney driving cycle [11], Perth driving cycle [12] and Melbourne peak driving cycle in Australia [13] and the ARTEMIS driving cycle in Europe [14] have become well-established means of estimating emission inventories in regions worldwide. Furthermore, many Asian countries have recently developed driving cycles such as the Chennai driving cycle in India [15], Hong Kong driving cycle [16], Vietnam driving cycle [17], Khon Kaen driving cycle in Thailand [18], Singapore driving cycle [8] and Colombo driving cycle in Sri Lanka [3]. Almost in all of these driving cycles, four main steps can be identified such as route selection, data collection, cycle construction and cycle assessment.

2.1. Route selection

Basic parameters such as traffic flow characteristics, land use, road type, topography and population density [7, 19] are considered when selecting routes to represent a road network. The experience and knowledge of local traffic conditions are also essential for route selection [17, 20]. Travel speed distribution of local vehicles [21] and origin–destination (O–D) pattern [11] were analyzed prior to route selection.

2.2. Data collection

Two methods are widely used in data collection: the chase car method and the on-board measurement method [22]. A hybrid of these two methods can also be used to consider local traffic behavior [7]. Although each method has inherent advantages and disadvantages [7, 22], the chase car method has been proposed as the most economical method of data collection [23].

2.3. Cycle construction

Commonly used methods for cycle construction are microtrip-based cycle construction, segment-based cycle construction, modal cycle construction and pattern classification cycle construction [24]. Many driving cycles which were developed recently have used the microtrip and segment-based cycle construction methods [8, 25, 26]. Although each method has inherent advantages, disadvantages and limitations, the modal cycle construction method accurately captures regional driving modes and partitions the data into snippets, even in areas with less or no idling states, such as expressways [23]. However, the microtrip method cannot be used for expressways because the microtrips are divided by adjacent stops in the speed–time profile, including the leading period of idle [27].

2.4. Cycle assessment

Cycle assessment, which ensures that the developed driving cycle accurately represents the collected data, is a vital step in driving cycle development. Researchers have used a variety of target parameters to compare models with collected data. Some driving cycles have used less than 20 target parameters [8, 20, 25, 28], while others have used more than 40 parameters [29]. In cycle assessment, target parameters or population parameters are initially calculated using the collected data set. Then, the same parameters are calculated for the candidate cycles. The summation of the absolute difference of each candidate parameter to the respective population parameter is calculated and called it as the performance value (PV) [15, 20, 25, 30]. The candidate cycle with the minimum PV is typically selected as the driving cycle for the area of concern. The speed acceleration probability distribution (SAPD) or speed acceleration frequency distribution is also commonly considered when selecting representative candidate cycles [10]. Another method of cycle assessment divides the data into two groups, develops independent driving cycles and then compares the parameters of the developed cycles [28]. This method is similar to the K-fold cross-validation method, which is widely used for model validation in transportation research [31].

DOI: 10.1002/atr

3. METHODOLOGY

3.1. Route selection

This study developed an economical approach to collect on-road speed–time data in Colombo, Sri Lanka, considering travel activity patterns throughout the city using O–D survey data. Two major trip types were identified: intercity and intracity trips. Routes were selected for both trip types, and replications were avoided during data collection to minimize the cost for data collection which accounts more than 80% of the total cost of driving cycle development.

3.1.1. Route selection for intercity trips

Actual trip patterns of vehicles entering Colombo were identified prior to route selection. By analyzing the O–D data collected by University of Moratuwa in the year 2014 at 10 major entry points to the Colombo [32], 57 intercity routes were identified. Based on entering traffic flow volumes from each road due to intercity trips, a traffic flow was assigned into each road segment. Finally, the segments were ranked in ascending order of traffic flow volumes. Results showed that more than 70% of total vehicles travelled through O–D locations utilized 33 routes out of the 57 identified routes for intercity trips. Roads with less vehicle frequencies were eliminated from consideration because they have less impact on total emissions in the city. Routes with high vehicle frequencies and subsequent high impact on total emissions were selected as representative road segments for the intercity trip category.

3.1.2. Route selection for intracity trips

Route selection for intracity trips in Colombo required an alternative method due to lack of available corresponding O–D data and complexity of conducting such a survey. School, commercial and residential areas in Colombo were identified using land use patterns and population densities, resulting in the selection of trip generators and attractors in the smallest administrative units in Colombo district (known as GN divisions). In the morning, commercial and school areas become trip attractors, and residential areas become trip generators. At noon, school areas become trip generators, and residential and commercial areas become trip attractors. In the evening, commercial areas became trip generators, and residential areas became trip attractors. Routes that connected trip generators and attractors in the morning, noon and evening were identified and superimposed onto one map. Considering the high traffic flow volumes on superimposed network, 72 routes were selected, including major and minor routes that potentially represent actual intracity driving in Colombo. The map shown in Figure 1 depicts the O–D zones for the evening period and the road network.

3.1.3. Representative routes for Colombo

Because Colombo contains intercity and intracity trip types, many selected routes for each trip category overlapped in this study. In order to avoid duplication, one route map was prepared by superimposing selected routes in both trip categories. Routes were divided into links using nodes or physical junctions in order to minimize segment length, so the links could be used to reconstruct routes for the O–D, as shown in Figure 2.

Selected routes were further divided into five groups according to daily traffic (DT), as presented in Table I. The segmented data also provided a basis for identifying the proper weightage for these routes, as given in Table II. The governing factor of route weightage was traffic volume, which represents the contribution of the respective route to total emission in Colombo.

After boundaries for the road groups were established, another table was prepared to identify traffic flow variations in each group in a typical day. A typical day (24 h) was divided into seven segments with noticeable traffic variation: 12:00 AM to 7:00 AM was called morning off-peak (MOP); 7:00 AM to 9:00 AM was called morning peak (MP); 9:00 AM to 12:00 noon was called interpeak 1 (IP1); 12:00 noon to 3:00 PM was called school peak (SP); 3:00 PM to 5:00 PM was called interpeak 2 (IP2); 5:00 PM to 8:00 PM was called evening peak (EP), and 8:00 PM to 12:00 AM was called night off-peak (NOP). A route with the highest DT within the group was then selected as a point estimator to represent each group in order to avoid underestimation in each group. Its timely variation of traffic flow was a representative value for the group, as shown in Table II.

DOI: 10.1002/atr

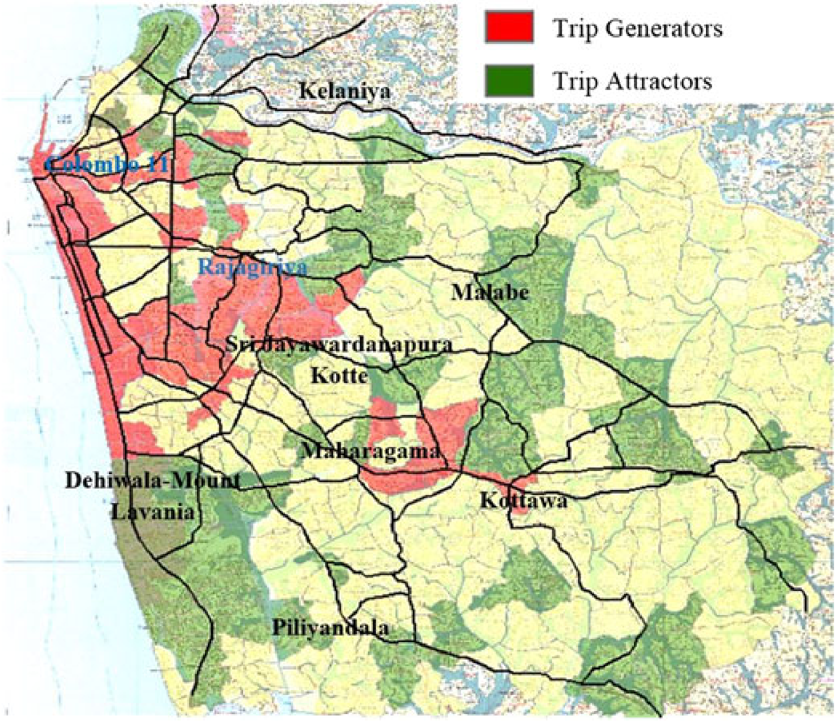

Figure 1. Trip attractors and generators in Colombo, Sri Lanka.

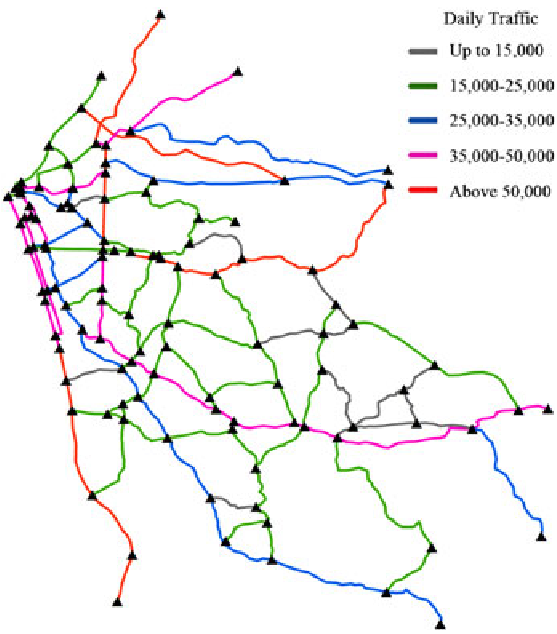

Figure 2. Selected route links for Colombo, Sri Lanka.

Although the sample size in Table II is ideal for the data collection, it is not practical or economical due to the high number of required trips. Therefore, a mathematically minimum sample, shown in Table III, was calculated by dividing all cells in Table II by its lowest traffic volume, which was 1100 in road group 2, MOP. This method reduces the required number of trips by reconstructing the O–D pairs using speed–time data in respective links, thereby reducing the cost of data collection.

DOI: 10.1002/atr

Table I. Group classification according to daily traffic.

Road group number	Total traffic volume (no. of vehicles)
1	Up to 15 000
2	15 001–25 000
3	25 001–35 000
4	35 001–50 000
5	Above 50 000

Table II. Daily traffic in road groups according to the time period of the day.

Road groups	MOP	MP	IP1	SP	IP2	EP	NOP
1	1725	3136	2060	1825	1237	2211	2275
2	1100	3968	3112	3265	2955	3172	2228
3	4441	7871	3652	3535	3723	8688	2310
4	4831	10 615	7829	5777	7878	9252	3725
5	12 372	14 687	9591	12 616	17 569	18 216	13 106

MOP, morning off-peak; MP, morning peak; IP1, interpeak 1; SP, school peak; IP2, interpeak 2; EP, evening peak; NOP, night off-peak.

Table III. Calculated minimum number of trips for data collection using average daily traffic.

Road groups	MOP	MP	IP1	SP	IP2	EP	NOP
1	2	3	2	2	1	2	2
2	1	4	3	3	3	3	2
3	4	7	3	3	3	8	2
4	4	10	7	5	7	8	3
5	11	13	9	11	16	17	12

3.2. Data collection

Consideration of traffic behavior in Colombo and budget availability led to the utilization of the on-board measurement method for data collection in this study. Five handheld GPS devices were used to record every 1 s speed data on road segments with ±3 m maximum estimated error. Because the proposed driving cycle was developed for light vehicles only, cars were used for data collection, and drivers with multidisciplinary education levels and professions were randomly selected to avoid data set bias. An attempt was made to maintain the collected data in proportion to the identified minimum sample in Table III. A total of 708 trips (trips for segments) were collected, equaling approximately 175 h as shown in Table IV.

4. CYCLE CONSTRUCTION

The Markov chain is a primary theory used in real-world driving cycle development [33, 34]. The modal cycle construction method, which is based on the Markov theory, was used with modification when developing a driving cycle for Sri Lanka.

Software called DCC2014 was developed using JAVA in order to handle a large data set. Also, it calculates transition probabilities and matrices every time when a data set is selected from the data

Table IV. Collected number of trips from each category according to the minimum requirement.

Road groups	MOP	MP	IP1	SP	IP2	EP	NOP
1	1	6	14	3	2	10	7
2	1	3	14	3	7	10	2
3	9	17	30	25	18	28	13
4	8	27	49	36	23	30	16
5	9	51	57	42	51	49	37

population. The software divided the data set into states, according to Table V, and assigned those data to corresponding bins. The software then calculated the probability of each state using Equation (1); results are shown in Table VI, column 1. A transition probability matrix of state changes was developed for the entire data set, and a new probability matrix for data selection was developed afterward using the method shown in Table VII. Data selection was conducted using the new probability matrix. The following sections present the descriptive methodology of cycle construction.

4.1. *Define the driving states*

Twenty-one states were defined for Colombo, using speeds and accelerations. Speeds in the collected data population were considered when identifying speed ranges for Colombo. Acceleration ranges

Table V. Defined states for transition probability matrix.

	Acceleration (ms^{-2})				
Speed (km/h)	$a \leq -0.8$	$-0.8 < a \leq -0.1$	$-0.1 < a \leq 0.1$	$0.1 < a \leq 0.8$	$a > 0.8$
$0 < v_m < 15$	State 1	State 2	State 3	State 4	State 5
$15 \leq v_m < 35$	State 6	State 7	State 8	State 9	State 10
$35 \leq v_m < 65$	State 11	State 12	State 13	State 14	State 15
$v_m > 65$	State 16	State 17	State 18	State 19	State 20

State 0: idle mode.

Table VI. Probabilities of each state.

	Column 1	Column 2	Column 3	Column 4	Column 5	Column 6
State	Initial probability of a state to get selected into the model (P_i)	Time allocated in driving cycle (s)	Time remaining from each state after selecting 100 s from state 0	Probability of get selected into the model (P_i) after selecting 100 s from state 1	Time remaining from each state after selecting 100 s from state 6	Probability of get selected into the model (P_i) after selecting 100 s from state 6
0	0.176	213	113	0.103	113	0.113
1	0.061	73	73	0.066	73	0.073
2	0.021	25	25	0.023	25	0.025
3	0.003	4	4	0.004	4	0.004
4	0.000	0	0	0.000	0	0
5	0.116	139	139	0.126	139	0.139
6	0.100	120	120	0.109	20	0.02
7	0.023	27	27	0.025	27	0.027
8	0.002	3	3	0.003	3	0.003
9	0.052	63	63	0.057	63	0.063
10	0.058	70	70	0.064	70	0.07
11	0.015	18	18	0.016	18	0.018
12	0.001	2	2	0.002	2	0.002
13	0.132	159	159	0.145	159	0.159
14	0.135	161	161	0.146	161	0.161
15	0.032	38	38	0.035	38	0.038
16	0.003	4	4	0.004	4	0.004
17	0.042	50	50	0.045	50	0.05
18	0.023	27	27	0.025	27	0.027
19	0.004	4	4	0.004	4	0.004
20	0.000	0	0	0.000	0	0

Column 1 shows the time proportion of each stage calculated for the entire data set. Column 2 shows the required length of teach state at the initial stage. Column 3 is the remaining duration of each state, assuming that 100 s data from the idle state were selected in the first step. Column 4 shows the new probabilities calculated for each state for the remaining durations using Equation (1). Assuming 100 s data were selected for the model from state 6 in the second step, new durations and new probabilities of each state are shown in columns 5 and 6.

DOI: 10.1002/atr

Table VII. New probability matrix for data selection.

	To		
From	State 0	State 1	State 2
State 0	0	$P_{0\to1}\times P_1$	$P_{0\to2}\times P_2$
State 1	$P_{1\to0}\times P_0$	0	$P_{1\to2}\times P_2$
State 2	$P_{2\to0}\times P_0$	$P_{2\to1}\times P_1$	0

were identified using previous research for developing driving cycle [33]. Considered speed and acceleration ranges are shown in Table V.

4.2. Calculate probability of occurrence of each state within driving cycle

Percentages for each state were calculated using the collected data set, as shown in Table VI, column 1. The length of the driving cycle for Colombo was determined to be approximately 1200 s due to the previously developed Sri Lankan driving cycle and economic constraints of running a developed cycle on a chassis dynamometer [3]. The length of each driving cycle state was calculated by multiplying 1200 by the proportion of each state, as shown in Table VI, column 2.

When selecting data for each state of the driving cycle, the required length of each state in Table VI, column 2, and the chance of getting selected of respective state in Table VI, column 1, had to be reduced. The probability of selecting each state was calculated using Equation (1), and then, new probabilities were calculated for each new selection. The new probability for the ith state P_i, after selecting the X_i time portion from the ith state is

$$P_i = \frac{[(1200\times p_i) - Xi]}{1200 - \sum_{i=0}^{i=20} Xi}, \tag{1}$$

where X_i denotes the time used from the ith state to construct the driving cycle and p_{i} denotes the initial probability of a state to get selected into the model. The cycle stopped when $\sum_{i=0}^{i=20} Xi = 1200$.

Table VI, column 4, shows the new probability of each state after selecting 100 s from state 0, and Table VI, column 6, shows the probabilities of each state after selecting 100 s from state 6.

4.3. New probability matrix for data selection

A transition probability matrix for state changes was constructed according to the literature [33, 34]. Because the driving behavior in Colombo did not fill each cell in the transition probability matrix (because some states did not occur), only higher probabilities were selected for the transition probability matrix for the Colombo driving cycle. Lower probabilities were not included in the driving cycle. A new matrix was defined by multiplying P_i (probability of each state as defined in Table V) and Equation (1) with $P_{i\to j}$ (probability of state change from ith state to ith state, as shown in Table VII). When calculating $P_{i\to j}$ (probability of state change), the data were first divided into 21 bins based on 21 defined states. Then, the number of occurrences of each state was calculated from the 21 states, and the probability of occurrence was calculated for each state from every other state. Software developed for driving cycle construction was used throughout the entire process.

When one mode was selected, the time allocated for the driving cycle from a respective state was reduced, leading to varying probabilities, as shown in Equation (1) and Table VI. It allowed other states to increase their probabilities, so those states get a higher chance of being selected for the driving cycle.

4.4. Chaining the data into driving cycle

State 0 was initially selected, and then, the state with the higher probability from state 0 was selected from the new probability matrix for data selection. Data was selected for the driving cycle until the length of the driving cycle reached 1200 s. For each selection, the new probabilities were calculated, and the new probability matrix for data selection was updated. The selected driving cycle is shown in Figure 3.

DOI: 10.1002/atr

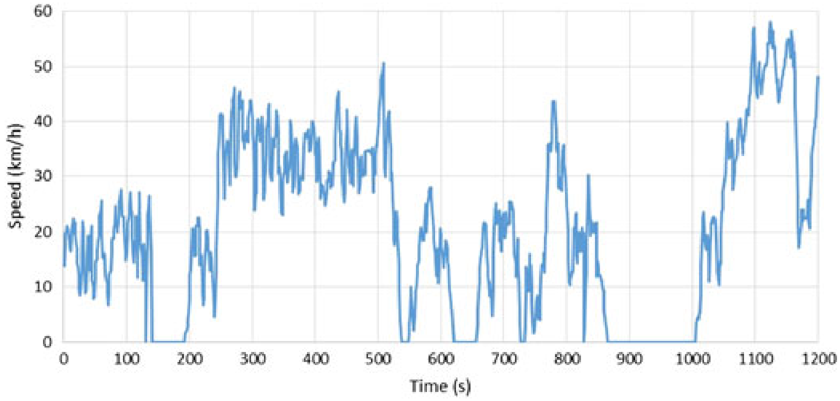

Figure 3. Colombo driving cycle.

Average speed and average running speed of the Colombo driving cycle were 20.3 and 28.8 km/h, respectively. Average acceleration and deceleration were 0.22 m/s^2. Percentages of time spent by each mode were 36.1, 30.65, 20.5 and 12.75% for acceleration, deceleration, idling and cruising modes, respectively. The driving cycle demonstrated 0.78 m/s^2 of root mean square acceleration (RMS) and 0.52 m/s^2 positive kinetic energy (PKE).

5. VALIDATION

Validation of the collected data set and the developed model was an essential component of this research. Statisticians have used various techniques to validate the developed models such as partitioning data set into two and calculating statistical parameters [35]. A similar method was used in the Pune driving cycle [28] in which the collected data were randomly divided into one-third (validation data set) and two-third (model building) data sets for analysis. Accordingly, target parameters were calculated for developed driving cycles using the complete data set, one-third data set and two-third data set. Ten parameters which were commonly used in emission-related driving cycles were selected for validating the developed driving cycle for Colombo, Sri Lanka as shown in the succeeding texts.

(1) Average speed of the entire driving cycle (V_A)
(2) Average running speed (V_R)
(3) Average acceleration (a_A)
(4) Average deceleration (d_A)
(5) Time proportions of driving modes for idling (P_{id})
(6) Time proportions of driving modes for acceleration (P_a)
(7) Time proportions of driving modes for cruising (P_c)
(8) Time proportions of driving modes for deceleration (P_d)
(9) RMS
(10) PKE

RMS and PKE are shown in Equations (2) and (3).

$$\text{RMS} = \sqrt[2]{\frac{1}{T}\left(\int_0^T a^2 dt\right)} \tag{2}$$

$$\text{PKE} = \frac{1}{dist}\sum_{i=2}^{n}\left\{v_i^2 - v_{i-1}^2 (\text{where } v_i > v_{i-1}) \text{ Else } 0\right. \tag{3}$$

After calculating the target parameters, PV is calculated for each parameter, and the results are shown in Table VIII. When developing driving cycle for Pune, India, it was identified that the

DOI: 10.1002/atr

Table VIII. Comparison of parameters.

Parameters	Population	Developed driving cycle	%Deviation to the population	One-third the data set	%Deviation to the population	%Deviation to developed driving cycle	Two-thirds of the data set	%Deviation to the population	%Deviation to developed driving cycle
V_A (km/h)	21.3	20.3	4.69	22.6	6.10	11.33	22.2	4.23	9.36
V_R (km/h)	27.2	28.8	5.88	29.9	9.93	3.82	28.6	5.15	0.69
a_A (ms^{-2})	0.2	0.22	10.00	0.21	5.00	4.55	0.21	5.00	4.55
d_A (ms^{-2})	0.2	0.22	10.00	0.21	5.00	4.55	0.21	5.00	4.55
P_a%	35.83	36.1	0.75	34.8	2.87	3.60	35.2	1.76	2.49
P_d%	31.53	30.65	2.79	31.7	0.54	3.43	30.4	3.58	0.82
P_{id}%	20.64	20.5	0.68	21.5	4.17	4.88	22.4	8.53	9.27
P_c%	12	12.75	6.25	12	0.00	5.88	12	0.00	5.88
RMS (m/s^2)	0.8	0.78	2.50	0.83	3.47	6.13	0.83	3.96	6.62
PKE (m/s^2)	0.47	0.52	10.64	0.48	2.96	6.94	0.50	6.38	3.85
Performance value of selected model			54.19						

maximum allowable deviation for the upper and lower limits were 15 and 5%, respectively, from the original parameters [28]. Same values were used when developing driving cycle for Colombo, and it can be seen that (in Table VIII) all the deviation with the population data set and driving cycle are in acceptable range. Even though PV is the only criterion for cycle validation in the past, recent driving cycles have also considered smallest sum square difference by calculating the summation of the difference between SAPD of the population and candidate cycle [23]. When calculating SAPD, the data set is initially divided into speed groups that are then divided into acceleration groups [12]. Similarly, this study defined 21 states using speed groups and acceleration groups. Because the cycle was constructed by considering SAPD, sum square difference was not used as an assessment parameter.

6. CONCLUSIONS

This paper describes a new, economical methodology for the development of driving cycles. This methodology is unique because the route selection criteria are simple but accurate, and it accurately represents actual driving patterns in studied areas. Because the routes were segmented using physical junctions or nodes, the required O–D patterns were easily reconstructed with a minimal number of data sets, thereby reducing the cost for data collection, which has a high cost component in driving cycle development. This data collection method can be modified to be more economical using the chase car and on-board methods according to method applicability to the considered area. The chase car method is applicable for areas with minimal and smooth traffic flows, but the on-board method can be used in areas with high, aggressive traffic flows. If the budget is not a constraint, accuracy of the collected data set can be improved by increasing the number of road groups by reducing the interval of each group or by increasing the sample size proportionate to the minimum sample size (Table IV).

This study used a method of cycle construction that utilized a Markov chain to develop transition probability matrix. A new matrix was then introduced to reduce the probability of being selected every time when a corresponding state was selected. This method enhanced the representativeness of developed driving cycle to the actual time proportion of each states defined using speeds and time (Table V) on the considered route network. Cycle duration was set to 1200 s, which sufficiently captured all 21 states in Colombo. However, further research is needed to select the optimum cycle length of Colombo, Sri Lanka, in order to reduce the cost of the dynamometer test.

ACKNOWLEDGEMENTS

Authors are thankful to Dr Thusitha Sugathapala for the guidance and motivation given and to Sustainable Energy Authority for the resources provided for successful completion of this research.

Further thanks go to Mr Sanjaya Liyanage and Ms Harini Gamage for helping develop the software for driving cycle development.

REFERENCES

1. Department of Motor Traffic. Total vehicle population (2007-2014) & new registration (2007–2014). 5/20/2015. http://www.motortraffic.gov.lk/web/images/stories/document/po14.pdf, Accessed 10/23, 2015.
2. Government of Sri Lanka. The National Environmental Act, no. 47 of 1980. *The Gazette of the Democratic Socialist Republic of Sri Lanka* 2008; **1557/14**(1): 1A–2A.
3. Gamalath, I, Fernando, C, Galgamuwa, U, Perera, L, Bandara, S. Methodology to develop a driving cycle for a given mode and traffic corridor; case study for Galle Road, Colombo, Sri Lanka. Proceedings of the Civil Engineering Research for Industry Symposium, Moratuwa 2012; 45–50.
4. Li T, Chen X Yan Z. Comparison of fine particles emissions of light-duty gasoline vehicles from chassis dynamometer tests and on-road measurements. *Atmospheric Environment* 2013; **68**: 82–91. DOI:10.1016/j.atmosenv.2012.11.031.
5. Ning Z, Wubulihairen M Yang F. PM, NOx and butane emissions from on-road vehicle fleets in Hong Kong and their implications on emission control policy. *Atmospheric Environment* 2012; **61**: 265–274. DOI:10.1016/j.atmosenv.2012.07.047.
6. Achour H, Carton JG Olabi AG. Estimating vehicle emissions from road transport, case study: Dublin City. *Applied Energy* 2011; **88**(5): 1957–1964. DOI:10.1016/j.apenergy.2010.12.032.
7. Tong H, Hung W. A framework for developing driving cycles with on-road driving data. *Transport Reviews* 2010; **30**(5): 589–615.
8. Ho S, Wong Y Chang VW. Developing Singapore driving cycle for passenger cars to estimate fuel consumption and vehicular emissions. *Atmospheric Environment* 2014; **97**: 353–362. DOI:10.1016/j.atmosenv.2014.08.042.
9. United State Environmental Protection Agency. Dynamometer driving schedules. 2/6/2013. http://www3.epa.gov/nvfel/testing/dynamometer.htm, Accessed 10/22, 2015.
10. Lin J, Niemeier DA. An exploratory analysis comparing a stochastic driving cycle to California's regulatory cycle. *Atmospheric Environment* 2002; **36**(38): 5759–5770. DOI:10.1016/S1352-2310(02)00695-7.
11. Kent JH, Allen GH Rule G. A driving cycle for Sydney. *Transportation Research* 1978; **12**(3): 147–152. DOI:10.1016/0041-1647(78)90117-X.
12. Lyons TJ, Pitts RO, Blockley JA, Kenworthy JR Newman PW. Motor vehicle emission inventory for the Perth Airshed. *Journal of the Royal Society of Western Australia* 1990; **72**: 67–74.
13. Watson, H. C., E. E. Milkins, and J. Braunsteins. Development of the Melbourne peak cycle. In *Second Conference on Traffic Energy and Emissions Melbourne*, Australia, 1982, pp. 1–21.
14. André M. The ARTEMIS European driving cycles for measuring car pollutant emissions. *Science of the Total Environment* 2004; **334–335**: 73–84. DOI:10.1016/j.scitotenv.2004.04.070.
15. Nesamani K, Subramanian K. Development of a driving cycle for intra-city buses in Chennai, India. *Atmospheric Environment* 2011; **45**(31): 5469–5476.
16. Tong HY, Hung WT Cheung CS. Development of a driving cycle for Hong Kong. *Atmospheric Environment* 1999; **33**(15): 2323–2335. DOI:10.1016/S1352-2310(99)00074-6.
17. Tong HY, Tung HD, Hung WT Nguyen HV. Development of driving cycles for motorcycles and light-duty vehicles in Vietnam. *Atmospheric Environment* 2011; **45**(29): 5191–5199. DOI:10.1016/j.atmosenv.2011.06.023.
18. Seedam A, Satiennam T, Radpukdee T Satiennam W. Development of an onboard system to measure the on-road driving pattern for developing motorcycle driving cycle in Khon Kaen City, Thailand. *IATSS Research* 2015; **39**(1): 79–85. DOI:10.1016/j.iatssr.2015.05.003.
19. Barrios CC, Domínguez-Sáez A, Rubio JR Pujadas M. Factors influencing the number distribution and size of the particles emitted from a modern diesel vehicle in real urban traffic. *Atmospheric Environment* 2012; **56**: 16–25. DOI:10.1016/j.atmosenv.2012.03.078.
20. Hung WT, Tong HY, Lee CP, Ha K Pao LY. Development of a practical driving cycle construction methodology: a case study in Hong Kong. *Transportation Research Part D: Transport and Environment* 2007; **12**(2): 115–128. DOI:10.1016/j.trd.2007.01.002.
21. Tamsanya, S., S. Chungpaibulpattana, and S. Atthajariyakul. Development of automobile bangkok driving cycle for emissions and fuel consumption assessment. In The 2nd Joint International Conference on "Sustainable Energy and Environment (SEE 2006)", Bangkok, Thailand, 2006, pp. 1–6.
22. Tong HY, Hung WT Cheung CS. On-road motor vehicle emissions and fuel consumption in urban driving conditions. *Journal of the Air & Waste Management Association* 2000; **50**(4): 543–554. DOI:10.1080/10473289.2000.10464041.
23. Galgamuwa U, Perera L Bandara S. Developing a general methodology for driving cycle construction: comparison of various established driving cycles in the world to propose a general approach. *Journal of Transportation Technologies* 2015; **5**(04): 191–203.
24. Dai, Z, Niemeier, D, Eisinger, D. Driving cycles: a new cycle-building method that better represents real-world emissions. Department of Civil and Environmental Engineering, University of California, Davis2008;
25. Amirjamshidi G, Roorda MJ. Development of simulated driving cycles for light, medium, and heavy duty trucks: case of the Toronto Waterfront Area. *Transportation Research Part D: Transport and Environment* 2015; **34**: 255–266. DOI:10.1016/j.trd.2014.11.010.
26. Seers P, Nachin G Glaus M. Development of two driving cycles for utility vehicles. *Transportation Research Part D: Transport and Environment* 2015; **41**: 377–385. DOI:10.1016/j.trd.2015.10.013.
27. Austin, TC, DiGenova, FJ, Carlson, TR, Joy, RW, Gianolini, K. Characterization of driving patterns and emissions from light-duty vehicles in California. Final Report 1993

28. Kamble SH, Mathew TV Sharma GK. Development of real-world driving cycle: case study of Pune, India. *Transportation Research Part D: Transport and Environment* 2009; **14**(2): 132–140. DOI:10.1016/j.trd.2008.11.008.
29. Zito R, Primerano F. *Drive Cycle Development Methodology and Results* < *br* />Transport System Center, University of South Australia: Australia, 2005.
30. Sigua R. Development of driving cycle for Metro Manila. *Journal of the Eastern Asia Society for Transportation Studies* 1997; **2**(6): 1995–2002.
31. Yanmaz-Tuzel O, Ozbay K. A comparative full Bayesian before-and-after analysis and application to urban road safety countermeasures in New Jersey. *Accident Analysis & Prevention* 2010; **42**(6): 2099–2107. DOI:10.1016/j.aap.2010.06.023.
32. Galgamuwa U, Perera L Bandara S. New methodology for developing driivng cycle(s) for Sri Lanka; case study, Colombo, Sri Lanka. *Journal of Society for Transportation and Traffic Studies (JSTS) Editorial Board* 2015; **6**(4): 37–45.
33. Shi Q, Zheng Y, Wang R Li Y. The study of a new method of driving cycles construction. *Procedia Engineering* 2011; **16**: 79–87. DOI:10.1016/j.proeng.2011.08.1055.
34. Bishop JDK, Axon CJ McCulloch MD. A robust, data-driven methodology for real-world driving cycle development. *Transportation Research Part D: Transport and Environment* 2012; **17**(5): 389–397. DOI:10.1016/j.trd.2012.03.003.
35. Neter J, Kutner MH, Nachtsheim CJ Wasserman W. *Applied Linear Statistical Models*Irwin: Chicago, 1996.